

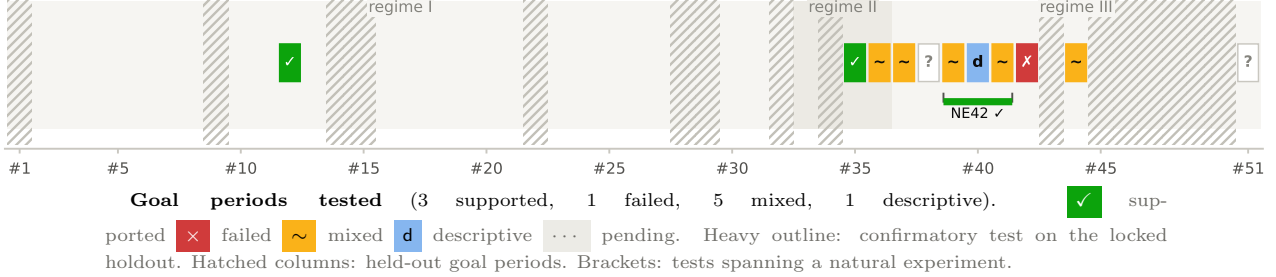
H26: The swarm is near-critical in what it says but subcritical in when it acts

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

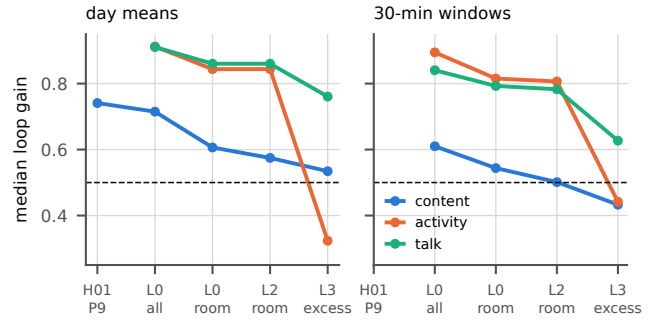
Bottom line. On corrected data, content and talk co-move inside rooms (gain ≈ 0.5) and activity barely does; no channel is near-critical, and H01's 0.74 is mostly drives.

level: hypothesis · pipeline checks 83% · updated 2026-10-04



WHAT HAPPENED

HH108 claimed that the swarm's *content* is near-critical (H01: $\beta J_0/n = 0.74$) while its *activity* is subcritical (loop gains ≤ 0.4). We put content, activity and talk on one estimand, $g = 1 - 1/VR$, at matched resolution (day means, 30-min windows). Field removal is nested and ends in a within-room minus cross-room contrast. We run 10 regime-III two-room units plus #35 and #51. In synthetic worlds, drives alone reproduce H01's number (0.57 from a global drive, 0.74 with more drive types). Round 1: content's room excess is genuine (median 0.53; null beaten in 6/10 units). Activity matches it (0.44 vs 0.43 at 30 min): **failed as posed**. Round 1b (fixed activity table, two embedding models): activity's room gain collapses to 0.12 at 30 min. Its cross-room ρ falls $0.50 \rightarrow 0.10$, the event-drop artifact. Content now exceeds activity (gap +0.41 [0.13, ...], 9/10 units). Talk matches content, and content stays ≈ 0.5 : **unresolved**. NE42: the room excess is channel-borne. The holdout is not run.



Observables. Round 1 (old activity table). Median loop gain over 10 regime-III two-room units, down the field-removal ladder. Steps: H01's estimator, whole-swarm (L0 all), room (L0), drive-projected room (L2), room excess (L3). Content (blue) falls from 0.74 to 0.53 (day) and 0.43 (30 min). On the old table, activity and talk sit above content at every step except day-level activity L3 (unpowered). Round 1b shows that the activity curve was the event-drop artifact (page 2).

PHYSICAL MODEL

Mean-field $O(n)$ soft spins and Curie–Weiss (11, 01). Spins: an agent's 32-d whitened statement mean (content), or its fraction of active or talking minutes. Coupling J acts within a room. Drives: global day and time-of-day, static goal fields, measured operator inputs and unmeasured room drives. C_{ab} : cross-agent covariance of drive-removed deviations. S_a : split-half signal variance. ρ_w, ρ_c : within- and cross-room correlation. N_r : room size. Order parameter: the room-excess loop gain g .

$$\rho_{w,c} = \frac{\sum_{\text{pairs}} \text{tr } C_{ab}}{\sum_{\text{pairs}} \sqrt{S_a S_b}}, \quad \rho_{\text{ex}} = \frac{\rho_w - \rho_c}{1 - \rho_c}$$

$$g = 1 - \frac{1}{1 + (N_r - 1) \rho_{\text{ex}}}, \quad g_{\text{eq}} = J, \quad g_{\text{day}} \rightarrow J(2 - J)$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Compare channels only at the same resolution and drive removal. Time-averaging alone turns an equal-time gain of 0.4 into 0.64, which made part of the HH108 gap. On corrected data, content and talk co-move within rooms (gain 0.4–0.6), and activity barely does (0.12 at 30 min). Treat a swarm-wide activity dip as a platform or schedule event, not as contagion. Operator messages explain none of the room content co-movement. No channel is near-critical: a gain of 0.5 means variance twice the independent level.

RESULTS VS PREDICTIONS

Prediction (locked)	Round 1 → 1b bge / gte	Verdict
P1a drives alone give H01 ≥ 0.3	0.57 (global); 0.67–0.74 (more drives)	passed
P2 L3 content median 0.25–0.5	0.53 → 0.52 / 0.47; ≥ 0.5 in 6 → 5 / 4 of 10	failed → met
activity L3 (day / 30 min)	0.32 / 0.44 → -0.38 / 0.12 (trimmed 0.16)	artifact
activity cross-room ρ (30 min)	0.50 → 0.10	artifact
P3 content – activity, 30 min	+0.03, 4/10 → +0.41 [0.13, ...] / +0.42, 9/10	failed → held
P3 content – activity, day	+0.32, CI $\ni 0$ → +0.76 / +0.42, CI $\ni 0$	failed
P3 content – talk, 30 min	-0.09 → -0.03 / -0.04	failed
P5 operator inputs $\leq 10\%$	≤ 0.01 in all units, both models	passed
P6 #51: content $\geq$ activity	0/4 → 1/4	failed
Card rule	failed (R2) → unresolved	changed
NE42 pseudo-rooms (native)	content, talk excess < 0 in merged #40, back at split	passed
#12 motion field (native)	on 0.72 vs off 0.66; removal halves it	passed / mixed

GOAL PERIODS IN DETAIL

G12	✓ 1b: supported (bge) / mixed (gte; N2b missed by 0.004)
G35	✓ 1b: supported
G36	~ 1b: mixed (bge) / failed (gte)
G37	~ 1b: mixed
G39	~ 1b: mixed (unchanged)
G40	d 1b: descriptive (unchanged)
G41	~ 1b: mixed (unchanged)
G42	× 1b: failed (unchanged)
G44	~ 1b: mixed (unchanged)
NE42	✓ 1b: supported (content and talk, both models; activity...)

+2 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

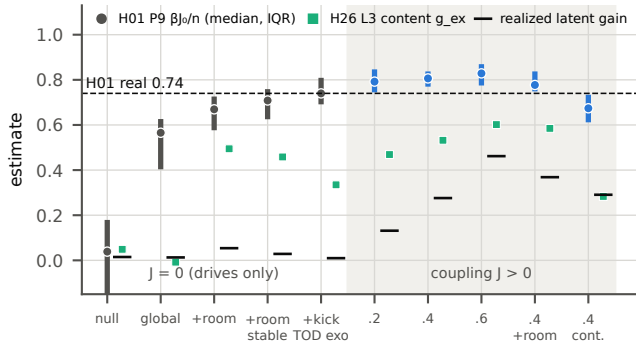
CAVEATS

Unmeasured room drives read as coupling: forks in #35, instructions in #38, the override in #44. Day-level activity is nearly unpowered (3–8 days), and the split-half variance of talk is unreliable. In synthetic worlds the estimator favors content, which strengthens the null. One-room units lack a baseline. Both embedding models agree, and the trimmed activity variant keeps the 30-min gap (+0.22 [0.01, ...]). Goals confound NE42, no cascade sizes exist yet (H34), and #51 confounds the post hoc size scaling.

ROUND 2

- **Essence.** whether ideas spread through a swarm more easily than effort does, measured so that channels are comparable. Round 1 says: not at matched resolution; the channels differ in where they co-move, not in how much.
- **H26-R1** Separate room coupling from room drive with the merge and split natural experiments.
- **H26-R2** Lagged, exposure-conditioned content gain.
- **H26-R3** Idea-cascade sizes (HH122) from H34.
- **H26-R4** Instrument checks: embedding swap, style control, H38 outages.

H26-content-near-critical · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H26-content-near-critical/README.md · ratings: H26 agent (suggested) (page 2)



Second figure. Synthetic validation: 24 worlds each, $N = 14$ in two rooms, 5 days. At zero coupling, H01’s P9 (gray) already reads 0.57–0.74 from drives and barely moves with J . H26’s L3 (green) tracks the realized latent gain (black) under global drives. It reads unmeasured room drives (“+room”) and imperfectly measured operator pulses as coupling.